

Detector-Plane Causality and Imprint Dynamics:

A Unified Measurement Framework for Quantum Coherence

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Abstract

We present a unified framework for quantum measurement that integrates particle-based field excitation, detector-plane causality, and coherence-resolved imaging into a single operational model. The framework combines three complementary components: (1) the Imprint Hypothesis, which treats quantum particles as primary field excitations and wave phenomena as contextual imprint propagation; (2) Detector-Plane Causality, formalized through the Quantum Measurement Stack (QMS) and QMCTB-01 benchmark, demonstrating that interference visibility is determined by detector-plane coherence preservation; and (3) Detector-Plane Imaging (DPI), which shows that spatial coherence structures evolve continuously at the detector plane prior to discretization.

Using nonlinear scalar-field simulations and detector-aware modeling, we demonstrate that interference patterns are encoded in pre-measurement coherence, decoherence is a continuous dynamical process, and measurement outcomes are causally determined by detector architecture. This framework reconciles quantum optics, statistical optics, and detector engineering within a QFT-consistent, experimentally testable model.

1. Introduction

Quantum measurement has historically been interpreted through probabilistic accumulation, observer-dependent collapse, and wave-particle duality. However, modern experiments reveal persistent ambiguities:

- Interference appears to “build up” over time
- Detection outcomes depend strongly on instrumentation
- Coherence and decoherence lack a unified physical description

This work resolves these issues by introducing a measurement-plane framework grounded in three principles:

1. Particle primacy — quantum objects are field excitations
2. Wave contextuality — wave behavior arises from field propagation
3. Detector causality — measurement outcomes are determined locally by detector systems

We unify these into a single operational structure spanning field dynamics, measurement physics, and detector-plane imaging.

2. The Imprint Hypothesis: Particle Primacy and Wave Contextuality

The Imprint Hypothesis establishes the ontological foundation:

- Quantum fields are the fundamental substrate
- Particles are discrete excitations created via field operators
- Wave-like behavior emerges from field propagation (Green's functions)

Key Principle

Wave phenomena (interference, diffraction, tunneling) are contextual imprints generated by particle-field interactions.

Measurement collapses the contextual imprint, fixing observable outcomes without negating particle existence.

3. Detector-Plane Causality: Measurement as a Local Process

Measurement is formalized through the Quantum Measurement Stack (QMS):

1. Field excitation and imprint formation
2. Detector-plane interaction
3. Data acquisition (DAQ)
4. Signal processing
5. Analysis

The detector plane defines the effective measurement boundary where information is irreversibly transformed.

QMCTB-01 Benchmark

Controlled double-slit simulations demonstrate:

- Identical propagation and photon statistics
- Detector-dependent outcomes

Results:

- Intensity-only detection → no interference
- Correlation-preserving detection → high-visibility interference
- No dependence on integration time

Core Conclusion

Interference visibility is determined by detector-plane coherence preservation, establishing detector-plane causality as the mechanism of measurement.

4. Detector-Plane Imaging (DPI): Coherence Before Discretization

DPI treats the detector surface as an imaging geometry that records evolving spatial coherence prior to discretization.

The detector records:

$$I(x, y, t) = \varphi^2(x, y, z_0, t)$$

This represents a continuous imprint field rather than discrete detection events.

Observations

Coherent regimes: - Concentric coherence rings

- Long-range spatial correlations
- Narrow spectral peaks

Decoherent regimes: - Fragmentation of coherence structure

- Spectral broadening
- Stochastic detector-plane textures

Key Insight

Interference patterns reflect real spatial structure present at the detector plane prior to discretization.

4.1 Oscillon–Photon Correspondence (Geometric Parallel)

A central structural insight underlying DPI is the geometric correspondence between nonlinear field oscillons and photon wavefront structure. This correspondence is **foundational**, not optional—it provides the bridge between classical field simulations and quantum-optical phenomena.

Oscillons in nonlinear scalar fields exhibit: - Spatial localization

- Long-lived phase coherence
- Envelope–carrier dynamics
- Stability under propagation with minimal dispersion

These properties directly mirror the **measured spatial and coherence structure of photon wavefronts** in interference experiments.

Crucially, this is **not an ontological identification**:

- Oscillons are classical nonlinear field excitations
- Photons are quantized excitations of quantum fields

Instead, the correspondence is **mathematical and geometric**:

Oscillons provide a classical solution class whose spatial and coherence structure is isomorphic to the imprint geometry generated by photon propagation.

This is why DPI—despite being purely classical—successfully reproduces:

- Interference ring structures
- Coherence-dependent fringe visibility
- Gradual interference buildup behavior

without invoking photon statistics or wavefunction collapse.

Within the Imprint Hypothesis framework, this correspondence has a precise interpretation:

- The photon (particle) generates a propagating field response (imprint)
- That imprint evolves according to field dynamics that are **well-approximated by oscillon-like solutions**
- The detector samples this structured imprint prior to discretization

Thus, oscillons function as a **mathematical parallel to photons at the level of field imprint geometry**, not as physical replacements. They provide a tractable classical proxy for studying coherence evolution, making DPI a valid pre-quantization diagnostic of measurement-plane structure.

This connection is essential: it explains how classical simulations can faithfully reproduce quantum-optical measurement morphology while remaining fully consistent with particle-based quantum field theory.

5. Coherence and Decoherence Dynamics

Coherence corresponds to sustained phase alignment and extended spatial correlations.

Decoherence arises from nonlinear mode coupling and energy redistribution, producing:

- Correlation-length decay $\xi(t)$
- Fourier spectral broadening
- Loss of structured spatial order

Decoherence is a continuous dynamical process rather than an abrupt transition.

6. Double-Slit Interference Reinterpreted

Using DPI-derived fields as illumination:

Coherent illumination: - High-visibility fringes

- Deep minima
- Structured diffraction envelopes

Decoherent illumination: - Reduced fringe contrast

- Lifted minima
- Suppressed interference structure

The double slit reveals coherence already present in the field rather than generating it.

7. Unified Measurement Model

1. Particle excitation as a field excitation
2. Imprint formation through propagation
3. Detector-plane sampling of the imprint
4. Detector-dependent filtering of correlations
5. Discrete observed outcome

Measurement is a causal filtering process applied to a pre-existing coherence structure.

8. Experimental Predictions

1. Detector dependence of interference visibility
 2. Exposure-dependent coherence length $\xi(t)$
 3. Local suppression of interference via detector-plane decoherence
 4. Pre-discretization spatial coherence structure
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9. Applications

- Quantum computing: detector-aware error correction
 - Quantum sensing: calibrated decoherence and noise characterization
 - Imaging systems: coherence-aware detector design
 - Semiconductor physics: tunneling as imprint-mediated transport
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10. Conceptual Implications

- Resolves wave-particle duality via particle primacy

- Defines collapse as a detector-plane process
 - Reinterprets interference buildup as statistical convergence
 - Establishes measurement as an instrumentation problem
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11. Conclusion

This unified framework establishes detector-plane causality and imprint dynamics as the operational basis of quantum measurement. By integrating particle excitation, field coherence, and detector physics, it provides a QFT-consistent and experimentally testable model that transforms quantum measurement from an interpretive problem into a controllable physical process.

References and Resources

Imprint Hypothesis: <https://doi.org/10.5281/zenodo.18383169>
QMCTB-01 Benchmark: <https://doi.org/10.5281/zenodo.18075090>
Code: <https://github.com/srikarr20/quantum-measurement-stack>